VULNER - Automated Network Scanning and Vulnerability Mapping

Project Overview

VULNER is an automated penetration testing and vulnerability mapping tool. The script receives a local network range and an output directory from the user, discovers active hosts, and performs controlled security checks against every detected target.

The user can choose between Basic and Full scan modes. Basic mode performs TCP and UDP scanning, service version detection, and weak credential checks. Full mode adds Nmap Scripting Engine vulnerability scanning and Searchsploit analysis.

All technical findings are saved into a structured results file. At the end of the process, the script displays the complete report, allows the user to search inside the results, and can compress the output directory into a Zip file.

Operational Logic Flow

1. Root Permission Check: The script checks whether it is running with root permissions. Root access is required for SYN and UDP scanning and for the controlled security tools used by the project.
2. Required Tools Check: The script verifies that Nmap, Hydra, and Zip are installed. If Full mode is selected, it also verifies that Searchsploit is available.
3. Network Input and Validation: The local IP address is detected automatically. The user enters a /24 network range, and the script verifies that the address is valid and belongs to the same local network.
4. Output Directory Preparation: The user enters a name for the output directory. The script validates the name, creates the directory and results file, or allows the user to overwrite an existing results file.
5. Scan Mode and Password List Selection: The user selects Basic or Full mode and chooses either the built-in password.lst file or a personal password list.
6. Username Collection and Host Discovery: Local Linux usernames are collected from /etc/passwd. Nmap discovers all active IP addresses in the selected network and saves them into an array.
7. Automated Target Scanning: Every detected IP address is scanned automatically. The selected stages run in sequence without additional user input between targets.
8. Final Results and Export: The report is displayed to the user, can be searched by a word or sentence, and can be saved together with the output directory inside a Zip file.

Scan Modes and Execution Logic

The project includes two scan modes: Basic and Full. Both modes are executed against every active IP address detected in the selected local network.

Basic mode performs the required network and credential checks. Full mode performs the same stages and adds vulnerability mapping through NSE and Searchsploit. The complete technical output is saved in results.txt, while progress messages are displayed in the terminal during each stage.

Scan Mode	Execution Logic	Result Handling
Basic	Runs a TCP SYN scan with service version detection, a controlled UDP scan, and weak credential checks for SSH, FTP, TELNET, and RDP when the relevant ports are open.	Saves all scan output and credential results into the report. Vulnerability mapping is marked as not executed because Basic mode was selected.
Full	Runs all Basic stages and adds an NSE vulnerability scan and Searchsploit analysis for detected open TCP services and versions.	Saves TCP, UDP, credential, NSE, and Searchsploit findings in separate report sections for every target.

Requirements Implementation

Network and output directory input

The script receives a network range and an output directory name from the user.

Basic or Full scan selection

The user can choose Basic or Full mode before scanning begins.

Input validation

IPv4 format, /24 CIDR, local network matching, directory name, menu choices, and password list paths are validated.

TCP and UDP scanning

Nmap scans every detected target for common TCP and UDP ports and performs service version detection.

Weak credentials

Hydra checks SSH, FTP, TELNET, and RDP login services when the related port is open.

Built-in or personal password list

The project creates a temporary built-in password.lst file and also allows the user to supply a valid personal list.

Vulnerability mapping in Full mode

NSE and Searchsploit run only when Full mode is selected.

Stage display and automation

Every stage is displayed in the terminal, and all detected targets are processed automatically.

Search and Zip output

The user can search inside results.txt and save the complete output directory into a Zip file.

Functions and comments

The script is divided into clear functions and includes comments explaining the main actions.

Tools Used in the Project

The project uses four main tools: Nmap, Hydra, Searchsploit, and Zip. Nmap is responsible for host discovery, TCP and UDP scanning, service version detection, and NSE vulnerability checks. Hydra is used for controlled weak credential testing. Searchsploit maps detected service names and versions to known exploit references, and Zip compresses the complete output directory.

Function / Stage	Purpose	Tool Used	Explanation
discover_active_hosts	Host discovery	Nmap	Scans the selected /24 network and saves all active IP addresses.
run_tcp_scan	TCP and service scan	Nmap	Scans the top 1000 TCP ports and detects service versions.
run_udp_scan	UDP and service scan	Nmap	Scans the top five UDP ports with controlled retry and timeout settings.
test_login_service	Weak credentials	Hydra	Tests collected usernames against the selected password list for supported login services.
run_nse_scan	Vulnerability mapping	Nmap NSE	Runs the vuln script category against the target in Full mode.
run_searchsploit	Exploit reference search	Searchsploit	Searches detected open service names and versions for matching exploit references.
create_results_zip	Results export	Zip	Compresses the complete output directory into a Zip file.

Creativity

One improvement added to the project is local network validation. The script does not accept an unrelated network range. It compares the entered /24 network with the machine's local network and asks again until the input is valid.

Another improvement is flexible result management. The script validates the output directory name, safely handles an existing directory, creates a structured results.txt file, allows the user to search inside the findings, and can compress the complete directory into a Zip file.

The project also includes controlled scan settings. UDP scanning uses limited ports, retries, and a host timeout to prevent long or hanging scans. Weak credential checks run only when a supported login port is detected as open, reducing unnecessary tests.

In addition, the script collects valid local Linux usernames automatically and checks each username separately. This produces a clear FOUND, NOT FOUND, or TEST ERROR result for every tested login service.

Conclusion

VULNER successfully automates network discovery, TCP and UDP service scanning, weak credential testing, and vulnerability mapping across the selected local network. The project validates user input, processes every detected target automatically, saves structured technical results, supports search inside the report, and exports the complete output into a Zip file. The separation between Basic and Full modes allows the user to perform either a focused scan or an extended vulnerability assessment.

Project Output Screenshots

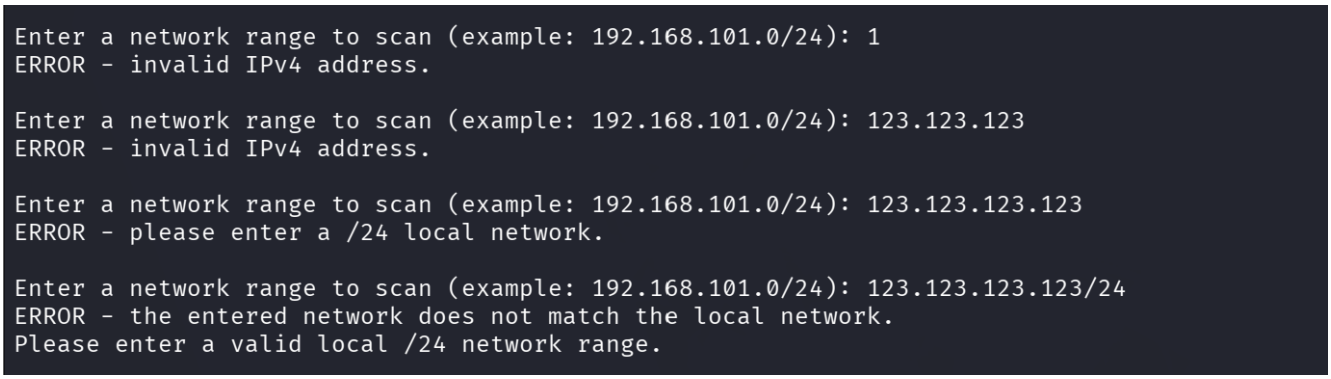
Startup, Root Permission and Required Tools Check

This screenshot shows the beginning of the script execution. The program verifies that it is running with root permissions and checks that the required tools are installed. The results confirm that Nmap, Hydra, and Zip are available before the scanning process begins.



Invalid Network Input Validation

This screenshot demonstrates how the script handles invalid network input. The user enters several incorrect values, including an incomplete IP address, an address without a CIDR range, and a network that does not match the local network. The script displays a clear error message after each invalid input and allows the user to try again.



Valid Network Range Selection

This screenshot shows a valid local network range being entered by the user. The script identifies the local IP address and local network, verifies that the entered address belongs to the same network, and saves it as the correct /24 network range.

```
Finding local IP address ...
Local IP address - 192.168.101.130
Local network - 192.168.101.0/24

Enter a network range to scan (example: 192.168.101.0/24): 192.168.101.130/24

Selected network range - 192.168.101.0/24
```

Existing Output Directory Handling

This screenshot shows how the script handles an output directory that already exists. The user can choose to overwrite the existing results file or enter another directory name. The screenshot confirms that the selected results file is prepared successfully.

```
Enter a name for the output directory: text
Validating output directory name ...
Output directory already exists.
1) Overwrite the existing results file
2) Enter another directory name

Choose option 1-2: 1

Existing results file was overwritten.
Results will be saved in - text/results.txt
```

Basic Scan Mode and Password List Selection

This screenshot shows the user selecting the Basic scan mode and the built-in password.lst file. Basic mode includes TCP and UDP scanning, service version detection, and weak credential checks.

```
Choose scan mode:
________________________________________
1) Basic - TCP, UDP, service versions and weak credentials.
2) Full - Basic scan, NSE and Searchsploit vulnerability mapping.

Choose scan mode 1-2: 1
Selected scan mode - Basic

Choose password list:
________________________________________
1) Use the built-in password.lst file.
2) Use my own password list file.

Choose password list 1-2: 1
Built-in password.lst file selected.
```

Username Collection and Active Host Discovery

This screenshot shows the script collecting local Linux usernames from the operating system. It then scans the selected network, identifies the active hosts, and displays all detected IP addresses that will be used as scan targets.

```
Choose password list 1-2: 1
Built-in password.lst file selected.

Collecting local usernames from the operating system...
Usernames selected for weak credentials checks:
──────────────────────────────────────────
[-] root
[-] alisa

Scanning the network for active IP addresses...
Network range to scan - 192.168.101.0/24
Network scan completed.
4 active IP addresses were found.

Active IP addresses:
──────────────────────────────────────────
192.168.101.1
192.168.101.2
192.168.101.254
192.168.101.130
```

Basic Scan Results for a Detected Target

This screenshot shows the complete Basic scan results for one detected target IP address. The report includes the TCP scan, UDP scan, weak credential handling, and vulnerability mapping status. The results show that the target was scanned for TCP and UDP services, that closed login services were skipped, and that NSE and Searchsploit were not executed because Basic scan mode was selected.

```
══════════════════════════════════════════
TARGET IP: 192.168.101.130
══════════════════════════════════════════

TCP SCAN RESULTS
──────────────────────────────────────────
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-19 06:28 -0400
Nmap scan report for 192.168.101.130
Host is up.
All 1000 scanned ports on 192.168.101.130 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 202.05 seconds

UDP SCAN RESULTS
──────────────────────────────────────────
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-19 06:31 -0400
Nmap scan report for 192.168.101.130
Host is up (0.00069s latency).

PORT    STATE  SERVICE     VERSION
123/udp closed ntp
137/udp closed netbios-ns
138/udp closed netbios-dgm
161/udp closed snmp
631/udp closed ipp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds

WEAK CREDENTIALS RESULTS
──────────────────────────────────────────
SSH port 22 is closed - credential check skipped.

FTP port 21 is closed - credential check skipped.

TELNET port 23 is closed - credential check skipped.

RDP port 3389 is closed - credential check skipped.

VULNERABILITY MAPPING
──────────────────────────────────────────
Not executed - Basic scan mode was selected.
```

Full Scan Results for a Detected Target

This screenshot shows the Full scan results for one detected target IP address. The report includes the TCP scan, UDP scan, weak credential handling, NSE vulnerability scan, and Searchsploit analysis. The results show the detected port states and services, indicate that closed login services were skipped, and show the detected vulnerability and exploit analysis sections.

```
==========================================
TARGET IP: 192.168.101.130
==========================================

TCP SCAN RESULTS
------------------------------------------
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-19 09:05 -0400
Nmap scan report for 192.168.101.130
Host is up (0.000033s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.3p1 Debian 5 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.89 seconds

UDP SCAN RESULTS
------------------------------------------
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-19 09:05 -0400
Nmap scan report for 192.168.101.130
Host is up (0.000048s latency).

PORT    STATE  SERVICE     VERSION
123/udp closed ntp
137/udp closed netbios-ns
138/udp closed netbios-dgm
161/udp closed snmp
631/udp closed ipp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

```
WEAK CREDENTIALS RESULTS
------------------------------------------
Service: SSH
Port: 22
------------------------------------------
Username: root
Password: Not found
Status: NOT FOUND

Username: alisa
Password: 1234
Status: FOUND

FTP port 21 is closed - credential check skipped.

TELNET port 23 is closed - credential check skipped.

RDP port 3389 is closed - credential check skipped.

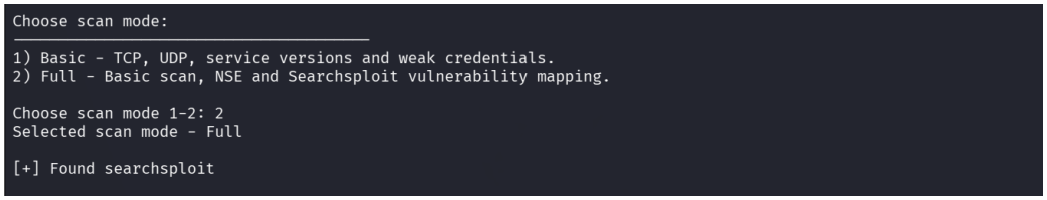
NSE VULNERABILITY RESULTS
------------------------------------------
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-19 09:05 -0400
Nmap scan report for 192.168.101.130
Host is up (0.000026s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.3p1 Debian 5 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:10.3p1:
|       CVE-2026-60002  9.4     https://vulners.com/cve/CVE-2026-60002
|       CVE-2026-60000  7.5     https://vulners.com/cve/CVE-2026-60000
|       CVE-2026-59999  7.5     https://vulners.com/cve/CVE-2026-59999
|       CVE-2026-60001  6.5     https://vulners.com/cve/CVE-2026-60001
|       CVE-2026-59998  6.5     https://vulners.com/cve/CVE-2026-59998
|       CVE-2026-59997  5.4     https://vulners.com/cve/CVE-2026-59997
|       CVE-2026-59996  5.4     https://vulners.com/cve/CVE-2026-59996
|_      CVE-2026-59995  5.4     https://vulners.com/cve/CVE-2026-59995
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.06 seconds
```

```
SEARCHSPLOIT RESULTS
------------------------------------------
Searching for: ssh ssh     OpenSSH 10.3p1 Debian 5 (protocol 2.0)
Exploits: No Results
Shellcodes: No Results
```

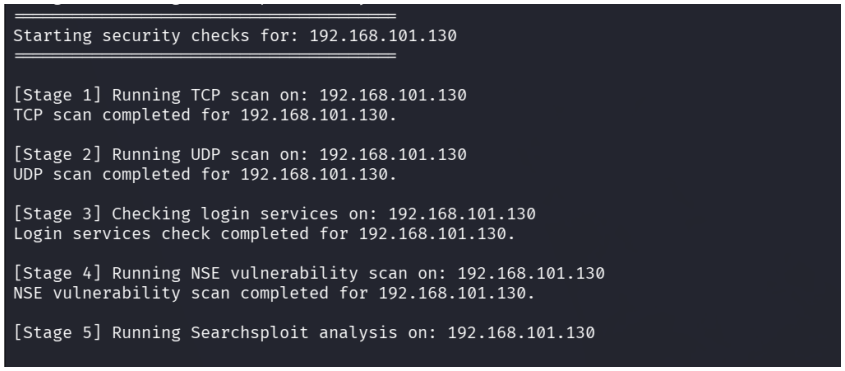
Full Scan Mode Selection

This screenshot shows the user selecting Full scan mode. Full mode includes all Basic scan stages and adds NSE vulnerability scanning and Searchsploit analysis. The script also confirms that Searchsploit is installed before continuing.



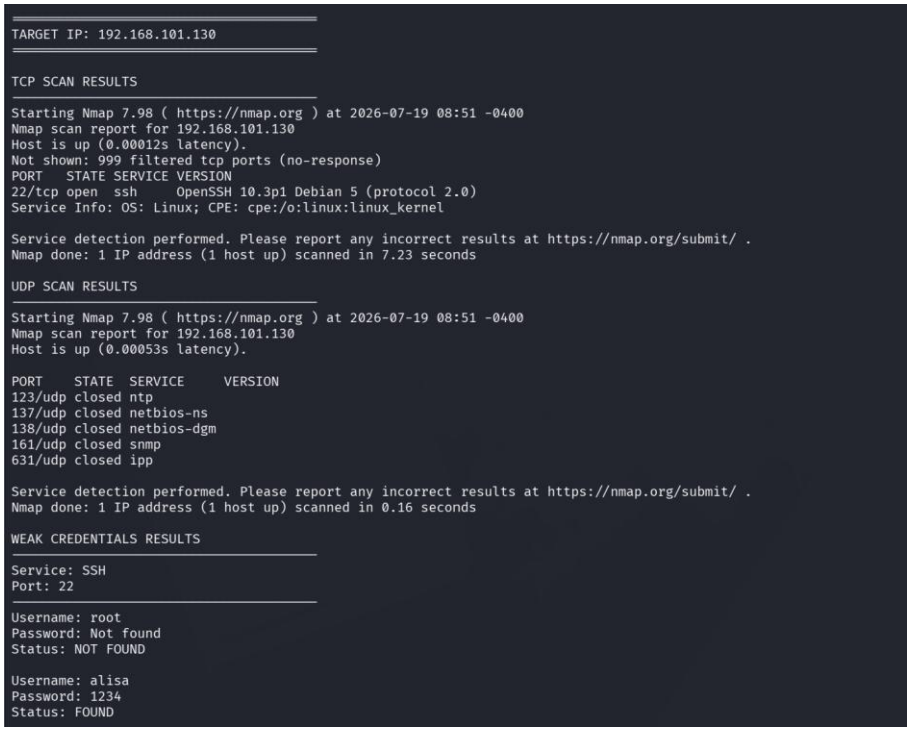
Full Scan Stages Execution

This screenshot shows the automated execution of the five stages included in Full scan mode. For the selected target, the script runs the TCP scan, UDP scan, login service checks, NSE vulnerability scan, and Searchsploit analysis in sequence without requiring additional user input between the stages.



SSH Weak Credentials Check

This screenshot shows the execution of the SSH weak credentials check on a target with port 22 open. The TCP scan identifies the SSH service and its version, and Hydra tests the collected usernames against the selected password list. The results show that no password was found for the root user, while weak credentials were successfully detected for the user alisa.



NSE Vulnerability Scan Result

This screenshot shows the NSE vulnerability scan performed on the detected SSH service. Nmap identifies the OpenSSH version and displays potential related vulnerabilities, including CVE identifiers and their severity scores.

```
NSE VULNERABILITY RESULTS
──────────────────────────────────────
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-19 09:05 -0400
Nmap scan report for 192.168.101.130
Host is up (0.000026s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.3p1 Debian 5 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:10.3p1:
|       CVE-2026-60002  9.4     https://vulners.com/cve/CVE-2026-60002
|       CVE-2026-60000  7.5     https://vulners.com/cve/CVE-2026-60000
|       CVE-2026-59999  7.5     https://vulners.com/cve/CVE-2026-59999
|       CVE-2026-60001  6.5     https://vulners.com/cve/CVE-2026-60001
|       CVE-2026-59998  6.5     https://vulners.com/cve/CVE-2026-59998
|       CVE-2026-59997  5.4     https://vulners.com/cve/CVE-2026-59997
|       CVE-2026-59996  5.4     https://vulners.com/cve/CVE-2026-59996
|_      CVE-2026-59995  5.4     https://vulners.com/cve/CVE-2026-59995
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.06 seconds
```

Searchsploit Analysis for a Detected SSH Service

This screenshot shows the Searchsploit analysis performed on the detected open SSH service. The script uses the detected service name and version as the search value. In this case, Searchsploit completed the search but did not find matching exploits or shellcodes.

```
SEARCHSPLOIT RESULTS
──────────────────────────────────────
Searching for: ssh ssh     OpenSSH 10.3p1 Debian 5 (protocol 2.0)
Exploits: No Results
Shellcodes: No Results
```

Personal Password List Validation and Selection

This screenshot shows the user selecting a personal password list file. When an invalid file path is entered, the script displays an error message and asks the user to try again. After a valid file path is provided, the script confirms that the personal password list was selected successfully.

```
Choose password list:
──────────────────────────────────────
1) Use the built-in password.lst file.
2) Use my own password list file.

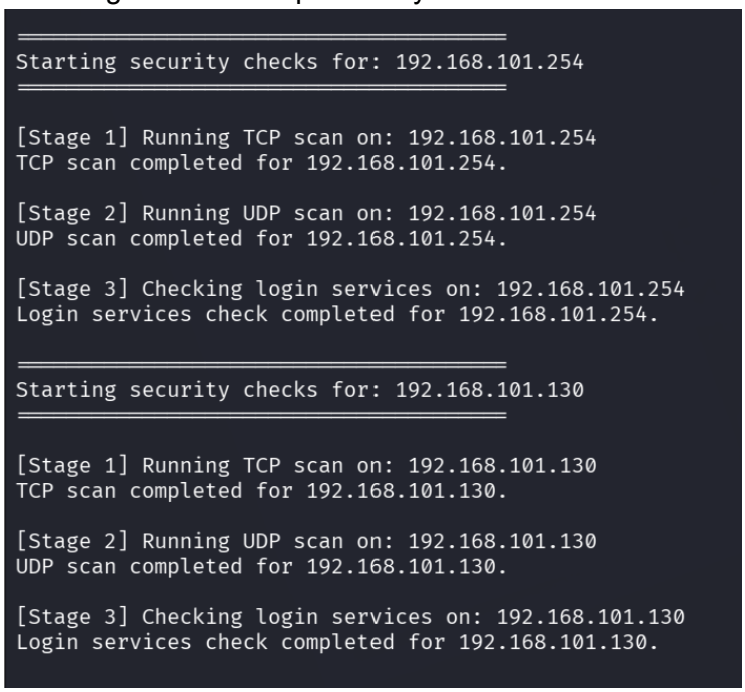
Choose password list 1-2: 2
Enter the full path to your password list file: lalala
ERROR - the selected password list is not valid.

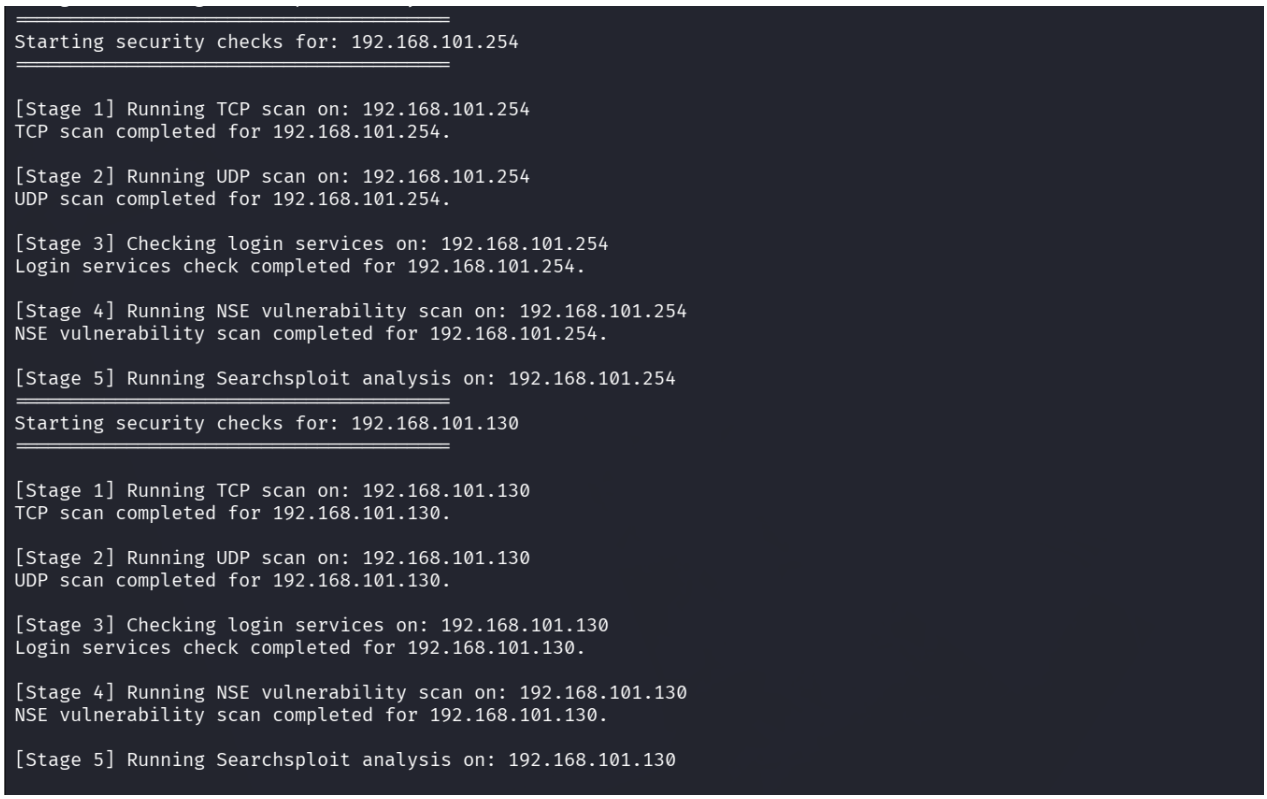
Choose password list:
──────────────────────────────────────
1) Use the built-in password.lst file.
2) Use my own password list file.

Choose password list 1-2: 2
Enter the full path to your password list file: /home/alisa/Documents/my_password_list.txt
Personal password list selected.
```

Automated Scan Across Multiple Targets

These screenshots demonstrate that the script automatically performs the selected security checks on multiple detected IP addresses. After completing the stages for one target, the program continues automatically to the next target without requiring additional user input. In Basic mode, it performs TCP scanning, UDP scanning, and login service checks. In Full mode, it also performs NSE vulnerability scanning and Searchsploit analysis.





Final Scan Report

This screenshot shows the final summary displayed at the end of the scanning process. It includes the scanned network, selected scan mode, number of active IP addresses found, location of the results file, and the completion message.

```
FINAL SUMMARY
──────────────────────────────────────
Network Scanned: 192.168.101.0/24
Scan Mode: Full
Active IP Addresses Found: 4
Results File: test/results.txt

VULNER Assessment Complete
```

Search Inside Results and Invalid Menu Input Handling

This screenshot shows the built-in results search feature and the script's handling of invalid menu input. When the user enters tcp instead of y or n, the program displays an error message and asks again. After a valid selection, the script searches the results file for the term tcp and displays the matching findings together with the related target IP addresses.

```
Would you like to search inside the results? (y/n): tcp
ERROR - enter y or n.

Would you like to search inside the results? (y/n): y
Enter a word or sentence to search: tcp

SEARCH RESULTS
──────────────────────────────────────
Target IP: 192.168.101.1
Result: TCP SCAN RESULTS

Target IP: 192.168.101.1
Result: Not shown: 1000 filtered tcp ports (no-response)

Target IP: 192.168.101.2
Result: TCP SCAN RESULTS

Target IP: 192.168.101.2
Result: Not shown: 999 closed tcp ports (reset)

Target IP: 192.168.101.2
Result: 53/tcp filtered domain

Target IP: 192.168.101.254
Result: TCP SCAN RESULTS

Target IP: 192.168.101.254
Result: Not shown: 1000 filtered tcp ports (no-response)

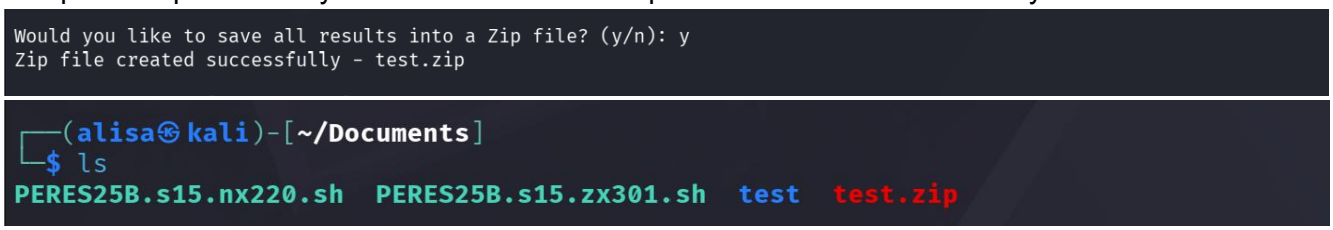
Target IP: 192.168.101.130
Result: TCP SCAN RESULTS

Target IP: 192.168.101.130
Result: Not shown: 1000 filtered tcp ports (no-response)

Search completed.
```

ZIP File Creation

This screenshot shows the user choosing to save the results into a Zip file. The script compresses the complete output directory and confirms that the Zip file was created successfully.



Run Without Sudo / Root Error

This screenshot shows the script being executed without root permissions and stopping with a clear error message. It also shows the program being restarted with sudo, after which the root permission check is completed successfully and the script continues.

